

Event-Driven Modeling for Short-Term Glucose Prediction from Irregular CGM-Insulin Data

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Abstract—Accurate short-term glucose forecasting is essential for effective management of type 1 diabetes. Many existing approaches preprocess continuous glucose monitoring (CGM) data using fixed-interval resampling, which can distort temporal relationships between glucose measurements and asynchronous therapy events. In this work, we propose an event-driven modeling framework that operates directly on irregularly sampled CGM–insulin pump data without imposing a fixed temporal grid. The formulation constructs prediction targets from observed measurements and supports unified modeling across a 30–60 minute forecasting window.

Using the DiaTrend dataset, we derive features capturing short-term glucose dynamics and therapy exposure. Experimental results show that a gradient-boosted model achieves a test RMSE of 23.26 mg/dL, corresponding to an 11.8% improvement over a naive baseline and outperforming reduced-feature and baseline models. Approximately 89.3% of predictions fall within Clarke Error Grid Zones A and B. Ablation analysis demonstrates that temporal history features provide the largest performance gains, while horizon encoding yields modest improvements. Additional comparisons with fixed-grid sequence models (LSTM), time-aware recurrent neural models, and a GRU-ODE-style baseline indicate that the proposed event-driven representation remains competitive with or superior to more complex neural architectures. These findings suggest that feature representation plays a more critical role than model complexity in glucose prediction from irregular clinical data.

Index Terms—glucose forecasting, continuous glucose monitoring, insulin pump data, irregular time series, event-driven modeling, horizon-aware modeling

I. INTRODUCTION

Accurate prediction of short-term blood glucose levels is critical for effective management of type 1 diabetes (T1D). Forecasting models can support insulin dosing decisions, reduce the risk of hypoglycemia and hyperglycemia, and contribute to automated insulin delivery systems [1]–[3]. The increasing availability of continuous glucose monitoring (CGM) data and integrated insulin pump records has enabled the development of data-driven approaches for modeling glucose dynamics [4]–[8].

Most existing glucose prediction pipelines for T1D rely on fixed-grid preprocessing, where CGM measurements are resampled to regular intervals prior to model training [2], [9]. While this simplifies modeling, it can smooth physiological dynamics and disrupt the temporal alignment between glucose measurements and asynchronous therapy events such as insulin boluses and carbohydrate intake. These limitations motivate modeling approaches that preserve the native timing of observations.

In addition, prior work typically formulates glucose prediction as a fixed-horizon task, training separate models for predefined forecasting intervals [1], [4], [10]. This approach limits the ability to capture shared predictive structure across nearby horizons and introduces redundancy in model training.

To address these challenges, we propose an event-driven modeling framework that operates directly on irregularly sampled CGM–insulin pump data. The proposed formulation constructs prediction targets from observed measurements and supports unified modeling across a 30–60 minute forecasting window by incorporating the realized prediction horizon (i.e., the time interval between the current observation and the target glucose measurement) as a model feature. More broadly, this work highlights that in irregular clinical time series, data representation can have a greater impact on predictive performance than increasing model complexity alone.

The contributions of this work are as follows:

- An event-driven formulation that preserves native measurement timing and avoids fixed-grid preprocessing.
- A feature representation capturing short-term glucose dynamics and therapy exposure from irregular data.
- A unified modeling approach across variable prediction horizons using horizon information as an input feature.
- A systematic empirical evaluation, including ablation analysis and statistical testing, to quantify the contributions of feature components.

- An empirical comparison demonstrating that the proposed event-driven representation remains competitive with or superior to both fixed-grid sequence models and continuous-time neural baselines.

II. RELATED WORK

Short-term glucose prediction has been widely studied using data-driven approaches applied to continuous glucose monitoring (CGM) data [1], [4]. Early methods relied on linear models and autoregressive techniques to capture temporal dependencies in glucose dynamics [1], [4]. More recent work has explored machine learning and deep learning approaches, including tree-based models, recurrent neural networks, and hybrid architectures, which have demonstrated improved predictive performance under controlled settings [1], [4]–[7], [10].

A common characteristic of many existing pipelines is the use of fixed-interval preprocessing, where irregularly sampled CGM measurements are resampled onto a uniform time grid prior to modeling [2], [9], [11]. While this simplifies model design, it can smooth short-term variability and disrupt the temporal alignment between glucose measurements and asynchronous therapy events such as insulin administration and carbohydrate intake. Several studies have noted that such preprocessing choices can influence both predictive accuracy and interpretability [9]. Modeling irregular time series without resampling has been studied in other domains using continuous-time and neural ODE approaches [12]–[15].

In addition to fixed-grid preprocessing, prior work often formulates glucose prediction as a fixed-horizon problem, training separate models for predefined forecasting intervals (e.g., 30 or 60 minutes ahead) [10]. While fixed-grid preprocessing affects how input data are represented in time, fixed-horizon formulations constrain how prediction targets are defined. Although this approach is straightforward, it limits the ability to share information across nearby horizons and increases the complexity of model deployment. Some recent efforts have explored multi-horizon prediction strategies [4], [10], but these approaches often still rely on discretized horizon definitions or fixed-grid representations.

Beyond temporal structure and horizon formulation, another important aspect of glucose prediction is the incorporation of contextual information, including recent glucose history, insulin dosing, and carbohydrate intake. Studies consistently show that short-term temporal features derived from recent measurements play a dominant role in predictive performance [2], [16]. However, the relative contribution of feature representation versus model architecture remains an open question, particularly in settings with irregular sampling.

In this work, we differ from prior approaches by adopting an event-driven formulation that operates directly on irregular CGM–insulin pump data without fixed-grid resampling. In addition, we encode the realized prediction horizon as a continuous feature, enabling unified modeling across a range of forecasting intervals. Our evaluation focuses on disentangling the contributions of feature representation, horizon

encoding, and model complexity through systematic ablation and statistical analysis.

III. DATASET AND EVENT-DRIVEN FORMULATION

We conduct our experiments using the publicly available DiaTrend dataset, which contains integrated records from continuous glucose monitoring (CGM) devices and insulin pump systems for individuals with T1D [16]. The full dataset includes data from 54 individuals across two cohorts, comprising over 27,000 days of CGM measurements and more than 8,000 days of insulin pump data. Because data availability varies across subjects, we focus on a subset of individuals with sufficient CGM, insulin, and carbohydrate records required for joint modeling of glucose dynamics and therapy events. After preprocessing and filtering for subjects with sufficient multimodal records, this results in a cohort of 17 subjects used in our experiments, as summarized in Table I.

Because these signals are generated by different devices and user interactions, the resulting data are not perfectly synchronized: CGM measurements may contain gaps or timing irregularities, while therapy events occur asynchronously relative to glucose measurements. This yields an irregular multimodal time series that is not naturally aligned to a fixed temporal grid [2], [9], [11], [13], [14].

Recent work on irregular time series has explored continuous-time modeling approaches, including time-aware recurrent networks and neural ordinary differential equation (ODE) frameworks such as ODE-RNN and GRU-ODE [13], [14]. These models explicitly account for irregular sampling by modeling latent state evolution between observations. While such approaches provide a principled framework for handling irregular data, they introduce additional modeling complexity and training challenges. In this work, we evaluate whether such complexity is necessary for short-term glucose prediction by comparing continuous-time neural models with simpler event-driven representations.

For each current timestamp t_{curr} (the time of an observed CGM measurement), we define a candidate prediction target by searching forward in time for the first future CGM measurement occurring at or beyond 30 minutes. If that future observation occurs within 60 minutes of t_{curr} , the pair is retained as a valid event-level sample. Let t_{target} denote the selected future timestamp. The realized prediction horizon is then defined as

$$h = t_{\text{target}} - t_{\text{curr}},$$

with $h \in [30, 60]$ minutes by construction. The prediction target for the sample is the glucose value observed at t_{target} .

To illustrate the event-driven target construction process, Figure 2 shows an example of how prediction targets are selected from observed CGM measurements.

As shown in Figure 2, the prediction target is defined directly from observed future measurements rather than from interpolated or fixed-interval resampled values. This ensures that each training sample reflects the true temporal spacing of the underlying data and avoids distortions introduced by fixed-grid preprocessing.

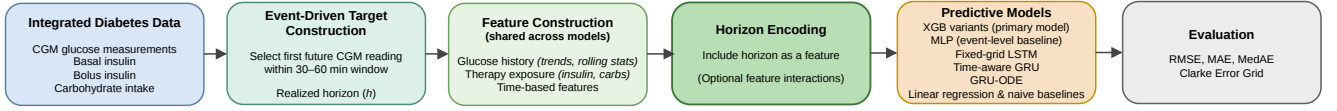


Fig. 1. Overview of the proposed event-driven glucose prediction framework. The pipeline operates directly on irregular CGM–insulin data without fixed-grid resampling. Prediction targets are constructed from observed future CGM measurements within a 30–60 minute window, and the realized prediction horizon is incorporated as a model feature.

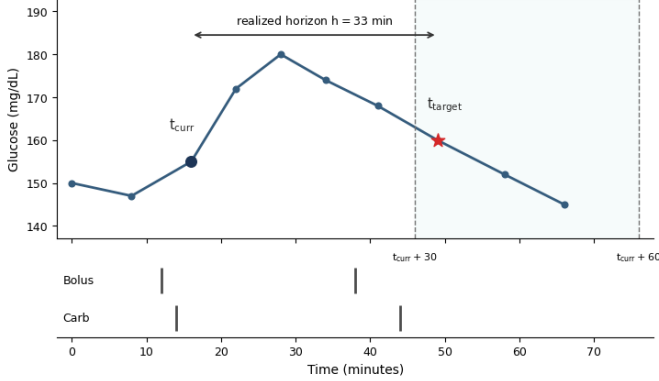


Fig. 2. Illustration of event-driven target construction. For a given current timestamp t_{curr} , the first future CGM measurement within the 30–60 minute window is selected as the prediction target. The time difference defines the realized prediction horizon h .

This event-driven construction produces a unified dataset spanning a continuous 30–60 minute forecasting window without requiring separate target-generation procedures for different horizons. It also ensures that the target always corresponds to an actually observed future glucose measurement. Figure 1 illustrates the overall framework.

The final event-level dataset is evaluated using subject-level partitioning so that the same individual does not appear in more than one split. This design supports evaluation on previously unseen subjects and reduces cross-subject information leakage. As discussed later in the Results section, the realized horizon distribution is highly imbalanced toward shorter intervals, with the majority of retained samples occurring near the lower end of the 30–60 minute window. This imbalance is an inherent consequence of the observed sampling structure and is therefore treated explicitly in the empirical analysis rather than removed through artificial resampling.

IV. FEATURE REPRESENTATION

For each event-level sample defined at a current timestamp t_{curr} , we construct a feature vector capturing glucose history, therapy exposure, and temporal context. The feature design emphasizes short-term glucose trends and variability derived directly from irregularly sampled observations, without requiring interpolation or resampling.

A. Glucose History Features

Short-term glucose dynamics are captured using CGM measurements observed prior to t_{curr} , including both the

most recent value immediately preceding t_{curr} and short-term history over preceding time intervals. These features summarize both absolute levels and local trends in glucose behavior. Specifically, we compute:

- Recent glucose values preceding t_{curr}
- Short-term differences (e.g., change over recent measurements defined over time intervals rather than fixed indices)
- Rolling statistics, including mean and standard deviation over recent time windows (in minutes)

These features provide a compact representation of local glucose trajectories and are computed directly from observed timestamps without resampling, ensuring that temporal relationships reflect the original irregular sampling structure.

B. Therapy and Event Features

To account for external interventions, we include features derived from insulin delivery and carbohydrate intake. These include:

- Aggregated basal and bolus insulin over recent time windows (in minutes)
- Carbohydrate intake events prior to t_{curr}
- Time since last insulin or meal event (in minutes)

These variables provide contextual information about recent therapy actions that influence future glucose levels.

C. Time-Based Features

Temporal context is incorporated through features describing the timing of each event. These include:

- Time of day (e.g., hour of day)
- Elapsed time since the previous CGM observation (in minutes)

Such features help capture periodic patterns and temporal variability in glucose behavior.

D. Horizon Feature

In addition to history and therapy features, we include the realized prediction horizon h as a model input. This enables a single model to operate across a continuous 30–60 minute forecasting window.

Importantly, horizon information is incorporated as a standard input feature rather than through explicit interaction terms. As shown in the Results section, including h provides modest improvements in predictive performance, while the majority of performance gains arise from history-based features. This observation is further supported by the ablation analysis presented in Section VI.

E. Feature Design Considerations

All features are constructed using only information available up to t_{curr} , ensuring that no future information is used during training. This design avoids information leakage and maintains consistency between training and inference.

Overall, the feature representation is designed to balance expressiveness and robustness, capturing short-term glucose patterns while remaining compatible with irregular sampling and event-driven target construction.

V. MODELING AND EXPERIMENTAL SETUP

A. Model Architectures

To evaluate the proposed event-driven feature representation, we consider multiple model classes with varying levels of complexity.

Our primary model is a gradient-boosted decision tree model (XGBoost) [17], which is well-suited for tabular data and has demonstrated strong performance in clinical prediction tasks.

To assess whether the benefits of the proposed representation generalize across model families, we additionally include a multilayer perceptron (MLP) operating on the same event-level features. This provides a simple neural baseline for evaluating the representation beyond tree-based methods.

We also include sequence-based neural baselines to compare against both fixed-grid and irregular-time modeling strategies. A fixed-grid long short-term memory (LSTM) model is trained on input sequences obtained by resampling CGM measurements to a fixed temporal grid, representing a conventional resampling-based pipeline. A time-aware gated recurrent unit (GRU) model incorporates elapsed-time information directly to account for irregular sampling [12]. Finally, we include a GRU-ODE-style model that evolves hidden states between observations, providing a representative continuous-time baseline for irregular time series modeling [14], [15].

Together, these models enable comparison between event-driven tabular methods, fixed-grid sequence models, and continuous-time neural approaches.

B. Model Variants

To isolate the contributions of different feature components, we define several XGBoost-based model variants:

- **XGB-Full**: Full feature set including history, therapy, time-based features, and horizon.
- **XGB-HN**: Horizon-aware model without explicit feature interaction terms.
- **XGB-NoH**: Model excluding the horizon feature.
- **XGB-Reduced**: Model using a reduced feature set that excludes glucose history features, retaining only therapy, time-based, and horizon features.
- **XGB-Full-W**: Weighted model that emphasizes longer-horizon samples during training.

These variants allow us to quantify the relative importance of temporal history features, horizon information, and sample weighting.

TABLE I

PARTITION SUMMARY AFTER EVENT-DRIVEN SAMPLE CONSTRUCTION.

Partition	Subjects	Samples	Mean target (mg/dL)	Target SD (mg/dL)
Train	12	271,694	163.59	58.17
Validation	2	32,645	171.24	61.86
Test	3	48,649	172.05	61.88

C. Training Procedure

All models are trained using subject-level data partitions to ensure that no individual appears in more than one split. This design supports evaluation on previously unseen subjects and reduces the risk of overfitting to subject-specific patterns.

For the event-driven tabular models, hyperparameters for XGBoost are selected based on validation performance. The MLP baseline is trained on the same event-level features using standard optimization settings with a fixed architecture and learning rate.

For sequence-based baselines, the fixed-grid LSTM is trained on these resampled input sequences, whereas the time-aware GRU and GRU-ODE-style models are trained on irregular event sequences using elapsed-time information. All models are trained using only information available up to t_{curr} , consistent with the event-driven formulation.

Random seeds are fixed where applicable to improve reproducibility across runs.

D. Evaluation Metrics

Model performance is evaluated using multiple complementary metrics commonly used in regression tasks [18]:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Median Absolute Error (MedAE)

To assess clinical relevance, we additionally evaluate predictions using the Clarke Error Grid [19], which categorizes predictions based on their potential clinical impact.

E. Statistical Evaluation

To assess the reliability of observed performance differences, we conduct statistical comparisons using bootstrap resampling and paired tests. Confidence intervals for RMSE are computed using bootstrap sampling on the test set. Paired comparisons between models are evaluated using both bootstrap-based tests and the Wilcoxon signed-rank test [20].

F. Implementation Details

All models are implemented in Python (version 3.12) using standard machine learning libraries. Feature construction, model training, and evaluation are performed using a consistent pipeline to ensure reproducibility.

TABLE II
MAIN PREDICTIVE PERFORMANCE ON THE TEST SET. RMSE
IMPROVEMENT IS COMPUTED RELATIVE TO THE NAIVE BASELINE.

Model	RMSE (mg/dL)	MAE (mg/dL)	MedAE (mg/dL)	RMSE improvement
XGB-HN	23.253	16.956	12.436	11.9%
XGB-Full	23.262	16.969	12.430	11.8%
XGB-NoH	23.299	16.997	12.462	11.7%
XGB-Full-W	23.912	17.496	12.916	9.4%
XGB-Reduced	25.799	19.100	14.368	2.2%
Linear Regression	25.809	19.093	14.327	2.2%
Naive	26.386	19.274	13.991	–

VI. RESULTS

A. Overall Predictive Performance

Table II summarizes predictive performance across the primary models. The strongest overall performance is obtained by the horizon-aware XGBoost variants, with *XGB-HN* achieving the lowest RMSE of 23.253 mg/dL and *XGB-Full* achieving a statistically comparable RMSE of 23.262 mg/dL. Both models substantially outperform *XGB-Reduced*, Linear Regression, and the Naive baseline. Relative to the Naive model, *XGB-Full* reduces RMSE by 11.8%, whereas *XGB-Reduced* improves RMSE by only 2.2%. These results indicate that the largest gains arise from history-based temporal features.

The MLP baseline (Table III) achieves competitive performance, outperforming *XGB-Reduced* and approaching the best XGBoost models. This suggests that the proposed representation benefits both tree-based and neural architectures, though it is most effectively leveraged by gradient-boosted models.

B. Comparison with Fixed-Grid, Event-Level Neural, and Continuous-Time Models

To evaluate whether more complex temporal models provide additional predictive accuracy, we compare the proposed event-driven approach with both fixed-grid baselines (*XGB-FixedGrid* and *LSTM*) and irregular-time neural baselines (*GRU* and *GRU-ODE*).

Table III summarizes the performance of these models. The fixed-grid LSTM baseline underperforms the event-driven XGBoost models, indicating that resampling-based sequence modeling does not provide advantages in this setting. The time-aware GRU model, which incorporates elapsed-time information directly, performs substantially worse, suggesting that naive handling of irregular sampling is insufficient for accurate prediction.

The GRU-ODE-style model improves over simpler sequence baselines such as the time-aware GRU and fixed-grid LSTM, demonstrating the value of explicitly modeling temporal dynamics between observations. However, it does not outperform the event-driven XGBoost models. These results indicate that the proposed feature representation captures the relevant temporal structure effectively, even when compared to more complex continuous-time neural architectures.

TABLE III
COMPARISON WITH FIXED-GRID AND CONTINUOUS-TIME NEURAL
MODELS ON THE TEST SET.

Model	RMSE (mg/dL)	MAE (mg/dL)	MedAE (mg/dL)
XGB-HN	23.253	16.956	12.436
XGB-Full	23.262	16.969	12.430
XGB-FixedGrid	23.440	17.120	12.650
MLP	23.388	16.833	12.124
LSTM (fixed-grid)	23.910	17.300	12.800
GRU (time-aware)	30.700	22.500	17.200
GRU-ODE	23.690	17.200	12.700

TABLE IV
ABLATION STUDY OF THE HORIZON-AWARE REPRESENTATION. NoH: NO
HORIZON FEATURE. HN: HORIZON-AWARE WITHOUT EXPLICIT
INTERACTION TERMS. FULL-W: WEIGHTED FULL MODEL.

Model	Horizon	History	Interactions
XGB-HN	Yes	Yes	No
XGB-Full	Yes	Yes	Yes
XGB-NoH	No	Yes	No
XGB-Full-W	Yes	Yes	Yes
XGB-Reduced	Yes	No	No
Linear Regression	Yes	No	No
Naive	No	No	No

Overall, these findings suggest that increasing model complexity alone does not guarantee improved performance, and that appropriate representation of irregular clinical data plays a more critical role.

C. Ablation Analysis

To isolate the contributions of the main representation components, Table IV compares models with and without horizon information, temporal history features, and explicit interaction terms. Removing history-based temporal features causes the largest degradation in performance, increasing RMSE from 23.262 mg/dL to 25.799 mg/dL. These results indicate that short-term glucose dynamics and recent therapy exposure provide the dominant predictive signal in this setting.

Removing the horizon feature yields a smaller increase in RMSE, from 23.262 mg/dL to 23.299 mg/dL, indicating that horizon encoding provides only a modest aggregate accuracy gain. This suggests that the practical value of horizon information lies more in supporting unified multi-horizon modeling than in substantially reducing average prediction error. Explicit horizon-feature interaction terms do not improve final performance, as *XGB-HN* slightly outperforms *XGB-Full*.

D. Statistical Reliability

Bootstrap confidence intervals and paired significance tests are reported in Tables V and VI. The confidence intervals of *XGB-Full* and *XGB-HN* overlap strongly, consistent with their nearly identical RMSE values. Paired bootstrap testing shows no significant difference between these two variants

TABLE V
BOOTSTRAP RMSE CONFIDENCE INTERVALS ON THE TEST SET.

Model	RMSE (mg/dL)	95% CI low (mg/dL)	95% CI high (mg/dL)
Naive	26.386	26.153	26.624
Linear Regression	25.809	25.593	26.048
XGB-Reduced	25.803	25.589	26.033
XGB-HN	23.261	23.036	23.482
XGB-NoH	23.306	23.077	23.530
XGB-Full	23.270	23.047	23.493
XGB-Full-W	23.913	23.683	24.135

TABLE VI
PAIRED STATISTICAL COMPARISONS AGAINST XGB-FULL.

Comparison	Bootstrap p	Wilcoxon p
Full vs NoH	0.0000	< 0.0001
Full vs HN	0.3413	n.s.
Full vs Reduced	0.0000	< 0.0001
Full vs Linear	0.0000	< 0.0001
Full vs Full-W	0.0000	< 0.0001

($p = 0.3413$), while both *XGB-Full* and *XGB-HN* significantly outperform *XGB-Reduced*, Linear Regression, and the weighted model. Although the difference between *XGB-Full* and *XGB-NoH* is statistically significant, the absolute performance gap is small, indicating limited practical impact of horizon encoding on aggregate RMSE. These results support the conclusion that temporal history features are essential, whereas explicit interaction terms do not provide additional benefit in the final experiments.

E. Clinical Evaluation

Figure 3 and Table VII summarize the Clarke Error Grid analysis for *XGB-Full*. Overall, 89.27% of predictions fall within Zones A and B, indicating clinically acceptable performance. Specifically, 78.79% of predictions fall in Zone A and 10.49% in Zone B. However, 6.76% of predictions fall in Zone D, indicating that clinically important errors remain non-negligible. Thus, while the overall clinical profile is favorable, the model should not be interpreted as fully robust in safety-critical conditions.

F. Performance Across Clinical Glucose Ranges

Table VIII reports RMSE stratified by target glucose range. The lowest error for *XGB-Full* is observed in the in-range region (70–180 mg/dL), where it achieves an RMSE of 20.142 mg/dL. Error increases substantially in hyperglycemia (26.715 mg/dL) and is highest in hypoglycemia (37.813 mg/dL). This pattern is consistent with the greater physiological variability and relative scarcity of low-glucose events [2], [16]. Importantly, horizon reweighting (assigning higher loss weights to samples with longer prediction horizons) does not improve performance in any range and slightly worsens hypoglycemia performance. Notably, the naive baseline achieves the lowest RMSE in the hypoglycemic range,

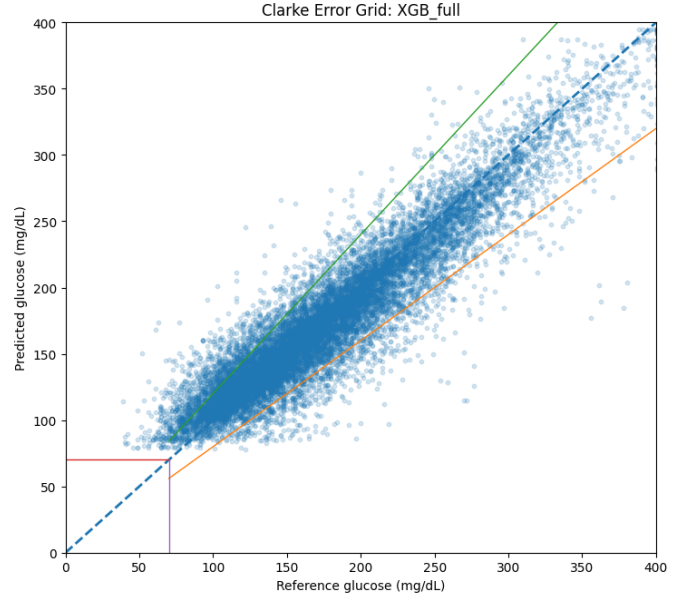


Fig. 3. Clarke Error Grid for the XGB-Full model on the test set. Most predictions fall within clinically acceptable Zones A and B.

TABLE VII
CLARKE ERROR GRID DISTRIBUTION FOR XGB-FULL ON THE TEST SET.

Zone	Count	Percent
A	38,328	78.785
B	5,103	10.489
C	1,930	3.967
D	3,288	6.759
E	0	0.000
A+B	43,431	89.274

indicating that all learned models remain limited in this safety-critical regime.

G. Performance Across Realized Horizons

The realized horizon distribution is highly imbalanced, with 47,656 of 48,649 test samples falling in the 30–35 minute range and only 187 samples in the 45–60 minute range. Table IX shows that error increases sharply with longer horizons. For *XGB-Full*, RMSE rises from 22.967 mg/dL at 30–35 minutes to 33.411 mg/dL at 35–45 minutes and 39.144 mg/dL at 45–60 minutes. These results indicate that overall performance is primarily driven by shorter-horizon forecasts due to the strong imbalance in the dataset.

Reweighting longer horizons does not improve performance; the weighted model performs worse overall and within each horizon bucket. This suggests that simple inverse-frequency weighting is insufficient to overcome the extreme scarcity of longer-horizon samples in the current dataset.

H. Error Analysis and Interpretation

Figure 4 shows the SHAP summary plot for *XGB-Full*. The most influential features are dominated by glucose his-

TABLE VIII
RMSE (MG/DL) ACROSS TARGET GLUCOSE RANGES. LR: LINEAR REGRESSION; RED: XGB-REDUCED; NOH: XGB-NOH; FULL: XGB-FULL.

Range	N	Naive	LR	Red	NoH	Full
Hypo (< 70)	408	33.183	37.469	37.494	37.435	37.813
In-range (70–180)	28,209	18.355	22.779	22.768	20.174	20.142
Hyper (> 180)	20,032	34.191	28.116	28.139	26.866	26.715

TABLE IX
RMSE (MG/DL) ACROSS REALIZED HORIZON BUCKETS. LR: LINEAR REGRESSION; RED: XGB-REDUCED; NOH: XGB-NOH; FULL: XGB-FULL.

Bucket	N	Naive	LR	Red	NoH	Full
30–35	47,656	26.185	25.580	25.572	23.001	22.967
35–45	806	34.406	37.028	37.003	33.790	33.411
45–60	187	40.183	43.137	42.949	39.403	39.144

tory variables, confirming the central role of recent glucose dynamics in prediction. Horizon-related variables contribute additional signal but are not among the dominant drivers of model performance. This is consistent with the ablation results, which show that history features matter substantially more than explicit horizon encoding in terms of aggregate RMSE. SHAP values are computed using the TreeSHAP method [21], which provides consistent feature attribution for tree-based models.

Figure 5 shows that the largest absolute errors are concentrated in extreme glycemic ranges, particularly low glucose values, and in longer-horizon cases.

VII. DISCUSSION

The results of this study highlight the importance of feature representation in short-term glucose prediction from irregular clinical data. Across all experiments, the largest improvements in predictive performance are driven by features capturing recent glucose dynamics and therapy exposure. In contrast, increasing model complexity alone yields relatively limited gains when the underlying feature representation is weak.

The comparison with neural baselines provides additional insight into the role of model complexity. While continuous-time models such as the GRU-ODE-style architecture capture irregular temporal dynamics more explicitly, they do not outperform the proposed event-driven approach. Moreover, the strong performance of XGBoost relative to both sequence-based and continuous-time models suggests that the proposed feature representation is effective across different model classes. The poor performance of the time-aware GRU further indicates that naive incorporation of elapsed-time information is insufficient, highlighting the importance of carefully designed feature representations.

The ablation analysis provides further insight into the role of different components of the proposed framework. Removing temporal history features leads to a substantial degradation in performance, confirming that short-term glucose trends

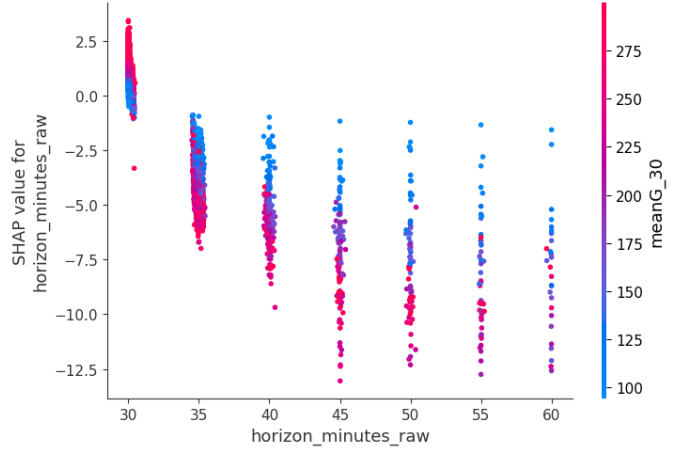


Fig. 4. SHAP summary plot for the XGB-Full model. Glucose history features dominate predictive importance, while horizon-related features provide smaller additional contributions.

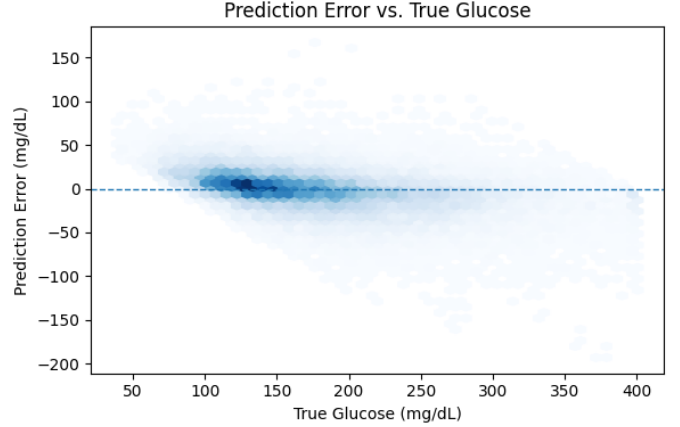


Fig. 5. Prediction error versus target glucose for the XGB-Full model. Darker regions indicate higher concentrations of predictions. Larger errors are concentrated in extreme glycemic ranges, particularly in hypoglycemia.

are the dominant predictors in this setting. Incorporating the prediction horizon as a feature yields modest improvements and enables unified modeling across a continuous forecasting window. However, explicit interaction terms between horizon and other features do not provide additional benefit in the evaluated models, suggesting that tree-based methods can capture relevant nonlinear relationships without requiring explicit feature interaction engineering (e.g., manually constructing interaction terms between horizon and other variables). The primary value of horizon encoding is enabling unified modeling across variable prediction intervals rather than improving aggregate accuracy.

The evaluation across glycemic ranges reveals that predictive performance is highest in the in-range region and degrades in both hypoglycemia and hyperglycemia. In particular, hypoglycemic events remain challenging to predict, which is consistent with their relative rarity and higher physiological variability. This limitation is also reflected in the Clarke Error

Grid analysis, where a non-negligible fraction of predictions fall into higher-risk regions. These findings indicate that, while overall predictive accuracy is strong, further improvements are needed to ensure reliability in safety-critical scenarios.

Analysis across realized prediction horizons shows a clear increase in error as the forecasting interval grows. However, the dataset is highly imbalanced toward shorter horizons, with the majority of samples concentrated near 30 minutes. As a result, aggregate performance metrics are primarily driven by shorter-horizon predictions. Attempts to correct this imbalance using simple reweighting strategies do not improve performance, suggesting that more sophisticated approaches or additional data may be required to improve long-horizon predictions.

This study has several limitations. First, the dataset size includes only 17 subjects, which may limit generalization across broader patient populations and should be addressed in future work with larger cohorts. Second, the distribution of prediction horizons is highly skewed, limiting the ability to learn robust patterns for longer horizons. Third, the evaluation focuses on a specific set of model classes and feature designs, and additional architectures or domain-specific features may yield further improvements.

Despite these limitations, the proposed event-driven formulation provides a flexible and conceptually simple approach for modeling irregular CGM–insulin data without imposing fixed-grid preprocessing. By preserving the native timing of observations, the framework avoids distortions introduced by resampling and enables consistent modeling across variable prediction intervals.

VIII. CONCLUSION

This work presents an event-driven modeling framework for short-term glucose prediction from irregular CGM–insulin data. By constructing prediction targets directly from observed measurements and avoiding fixed-grid preprocessing, the proposed approach preserves the native temporal structure of clinical data and enables unified modeling across a 30–60 minute forecasting window.

The results demonstrate that feature representation, particularly short-term glucose history, plays a more critical role than increasing model complexity alone. While horizon information enables flexible multi-horizon modeling, its contribution to aggregate accuracy is modest relative to history-based features.

Comparisons with fixed-grid and continuous-time neural models further show that the proposed event-driven representation remains competitive with more complex modeling approaches.

Future work will focus on improving predictive performance in hypoglycemia and at longer prediction horizons, where current models remain limited. Additional directions include leveraging larger and more diverse datasets and exploring modeling approaches that better capture rare and safety-critical events.

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